

Analyzing the Impact of Housing Supply, Demand, and Pricing on Property Sales in U.S. Metropolitan Areas

This project examines how variations in local housing **supply**, **demand**, and **pricing** influence property sales across major U.S. metropolitan areas.

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The Problem

- Understanding how supply, demand, and pricing affect property sales is critical in US Metropolitan Areas.
- Misinterpreting trends can lead to poor investment decisions or policy mistakes.
- Identifying which factors most influence sales allows for **more efficient resource allocation** and **targeted market strategies**.
- Insights can guide actions in **high-demand metros** and improve overall **housing market efficiency**.

Data Approaches

- Cleaned, merged and filtered data, focusing on relevant numerical features like supply, demand, pricing, and sales.
- Averaged data from 2019 to 2025
- Created groups (Top 10, Top 100, Others) to highlight differences among metro areas.
- Created **scatterplots** and a **correlation heatmap** to explore relationships between features and sales.

Dataset

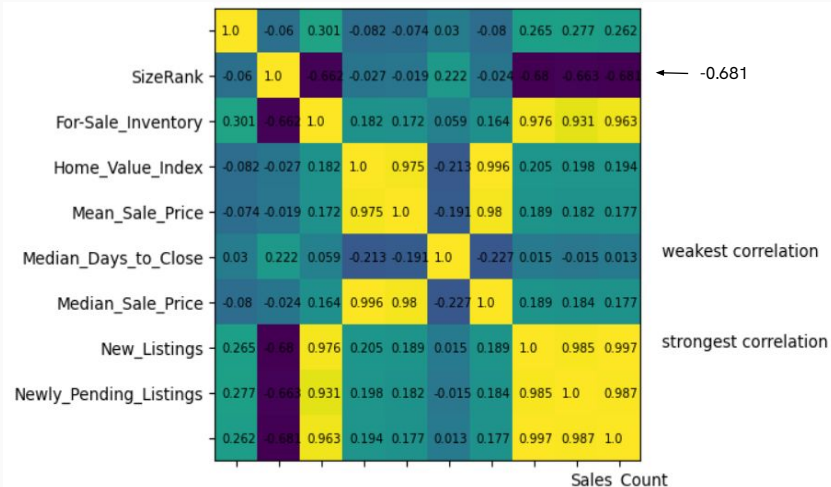
- Source: Zillow's Research Public Datasets
- Covers **100+ U.S. metropolitan areas** (MSAs) from **2019** to **2025**.

Features include:

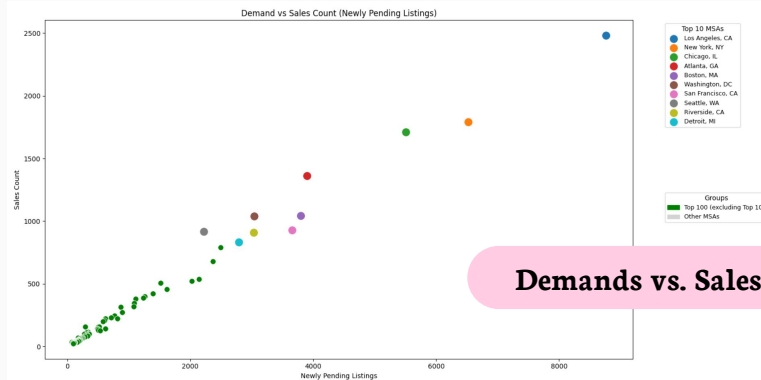
- Supply: **For-Sale_Inventory**, **New_Listings**
- Demand: **Newly_Pending_Listings**
- Pricing: **Median_Sale_Price**, **Mean_Sale_Price**, **Home_Value_Index**
- Market efficiency: **Median_Days_to_Close**
- Outcome: **Sales_Count**

Correlation Heatmap

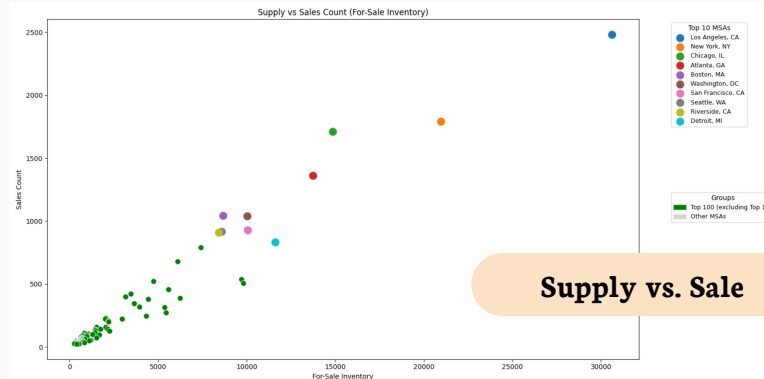
- The heatmap shows how numerical features relate to each other and to **Sales_Count**.
- **New_Listings** has the strongest correlation with sales, representing the effect of greater supply leading to increased sales. This is especially the case since **For-Sale_Inventory** and **Newly_Pending_Listings** have weaker, but similarly strong correlations.
- **Median_Days_to_Close** has the weakest correlation with sales, indicating that the amount of time spent waiting for a sale to be finalized is not a significant factor in determining whether the sale is made.
- It also interesting to note that **Mean_Sale_Price** and **Median_Sale_Price** have almost the same correlation with most other features and each other. This could suggest a potentially normal distribution of the data.



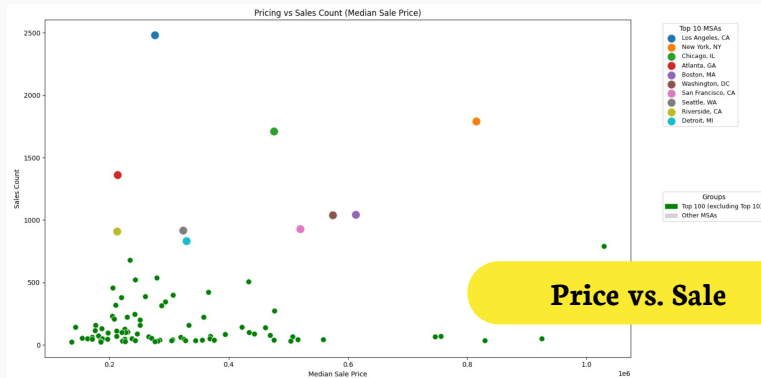
Insights



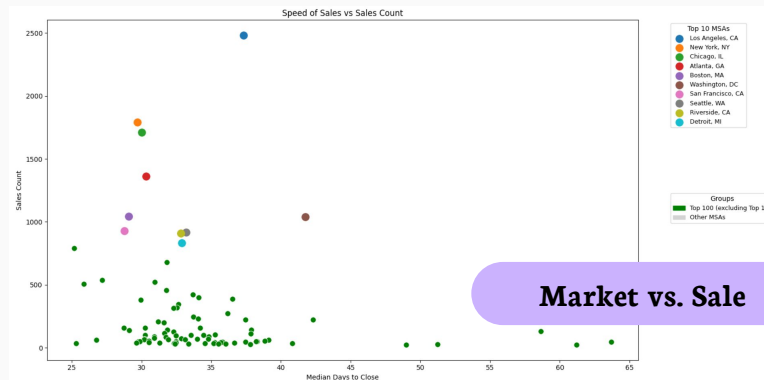
Demands vs. Sales



Supply vs. Sale



Price vs. Sale



Market vs. Sale

Conclusions

Key Takeaways

- **Housing demand** (pending listings) is the **strongest driver of property sales** in U.S. metropolitan areas.
- **Supply** has a similarly strong correlation to sales count, while **transaction speed** has the lowest overall impact.
- Median and Mean sale price have similar, not particularly significant, impacts on sales.
- **City size** (SizeRank) also has an **inherent role** in sales count, especially for metropolitan areas.

Recommendations

- Developers and policymakers should focus on **aligning housing supply with demand** to maximize sales efficiency.
- Monitoring median home prices helps anticipate sales trends and market affordability.
- Targeted strategies for top metro areas can have an outsized impact on overall market activity.
- Analyzing factors more indirectly related to the housing market, such as aggregated data of income, household size, etc. could prove valuable.